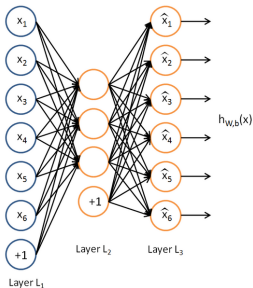


Overview

- Motivation for dimensionality reduction
- Feature selection
 - Wrappers
 - Filters
 - Embedded methods
- Feature transformation
 - Principal Component Analysis (PCA)
 - Autoencoders
 - Clustering

Autoencoders

- Unsupervised algorithm that tries to learn an approximation of the identity function $h_{\mathbf{W},\mathbf{b}}(\mathbf{x}) \approx \mathbf{x}$
- The middle layer of the autoencoder can be used as the transformed feature set



Overview

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Clustering

- Grouping of “similar” instances in the data sample
- Replacing a high dimensional data entry with a cluster label
- Deterministic clustering (e.g., K-Means) gives only one label per input
- Soft clustering gives probability of a sample belonging to each cluster
- More in the next lecture!

What have a learned so far

- Dimensionality reduction for visualization, compression, avoid curse of dimensionality
- Feature selection to select the most informative features
- Feature transformation to transform the features into a reduced space
- **Readings:** Alpaydin 6.1-6.3